Goniometrix

Wearable Smart Goniometer for Real-Time Range-Of-Motion Guidance

Decoding motion, one degree at a time.

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Jacobs Design Showcase

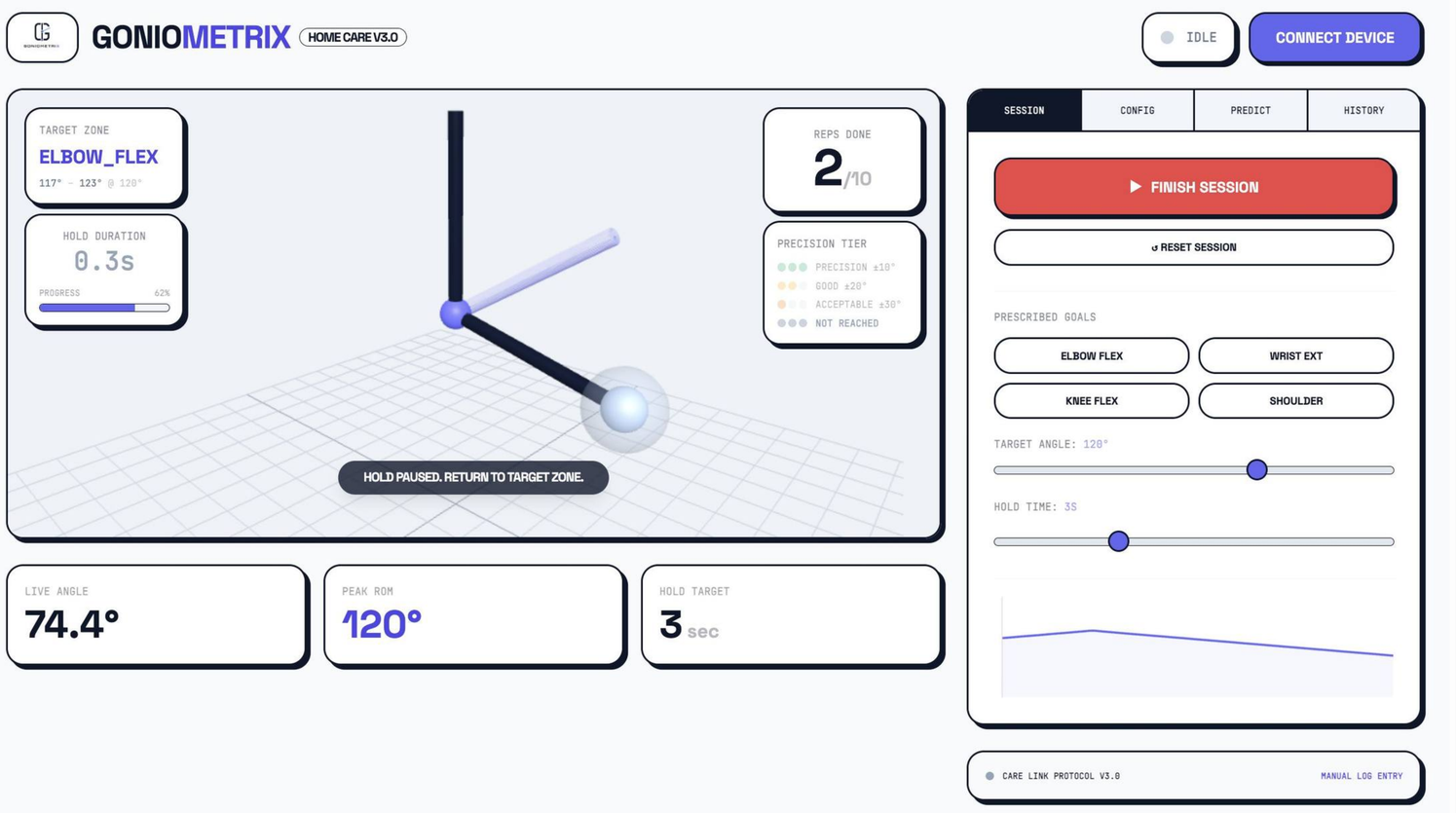
The Challenge

At-home rehab and movement training often rely on subjective self-assessment or clinician supervision. This makes joint-angle exercises hard to repeat consistently, difficult to monitor, and harder to trust over time.

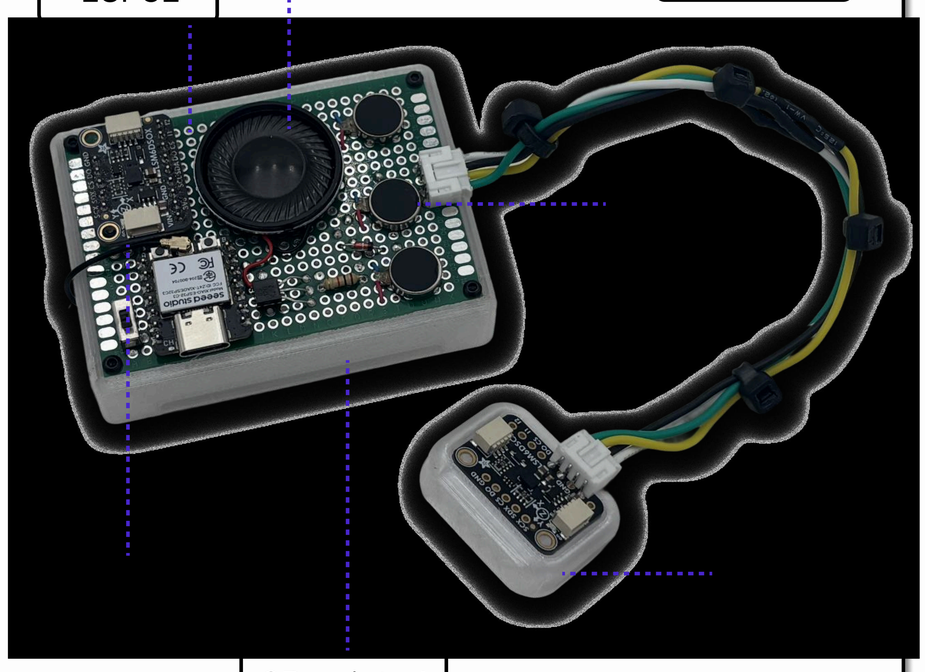
Our Solution

Goniometrix is a wearable smart goniometer that measures joint angle in real time and guides users toward target ROM goals using haptic and audio feedback. A companion GUI visualizes live movement, session performance, and progress over time.

Companion Interface



Final Prototype



How It Works

1 Sensing

Two IMUs capture joint orientation after calibration.

2 Prediction

ESP32 computes angle, classifies motion, and estimates target-angle progress from trends.

3 Actuation

Vibration motors and speaker deliver real-time corrective cues while the GUI logs session.

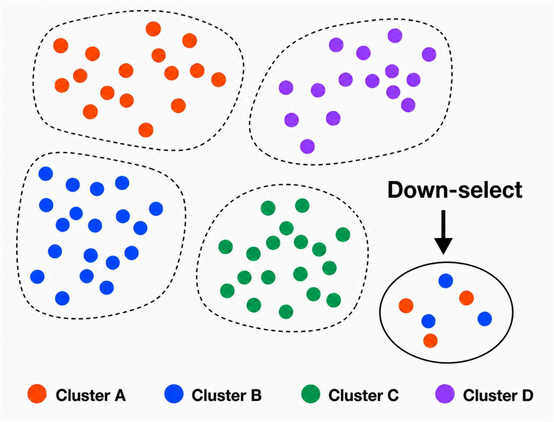
4 Intended Experience

A simple, non-visual feedback loop that supports independent, consistent, and controlled movement.

Design Journey

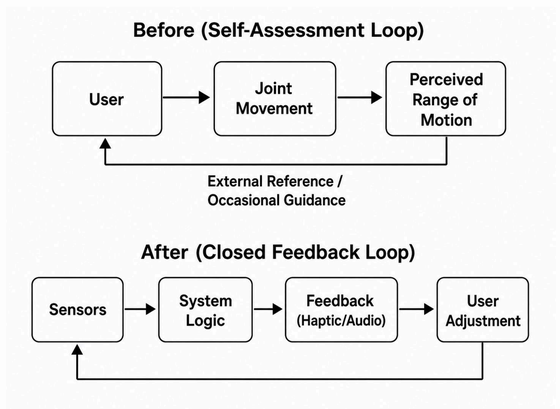
1. Concept Exploration

We explored multiple directions, clustered concepts, and compared them by feasibility, novelty, impact before selecting a wearable feedback-based direction.



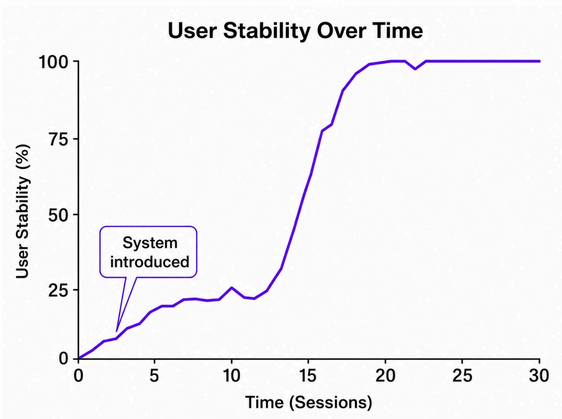
2. System Mapping

We mapped the system before and after intervention to understand feedback loops, trust, and long-term behavior.



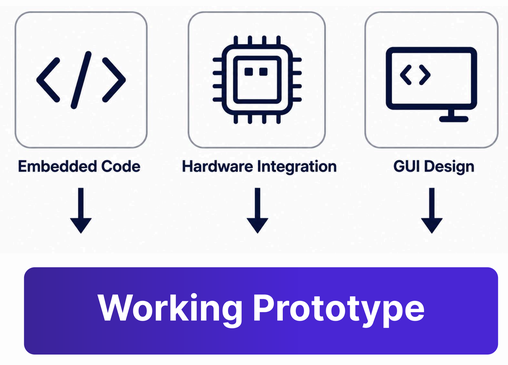
3. Behavior Over Time

User stability was identified as a key stock: it fluctuates early, improves with repeated feedback, and eventually plateaus as motion becomes more consistent.



4. Prototype Build

We translated the system into a working prototype spanning hardware, embedded code, and GUI design.



Key Insights

- User stability improves through a closed feedback loop.
- Accurate feedback builds trust and encourages continued use.
- Corrective motion decreases as movement becomes more repeatable.
- Session data enables progress tracking and future adjustment.

Potential Applications

- At-home rehabilitation
- Athletic training
- Ergonomic coaching

What We Built

- Wearable dual-IMU sensing system.
- ESP32-based real-time processing
- Haptic + audio feedback
- Desktop GUI with live angle, target zone, reps, and history.